

Sectoral Aggregation and Misallocation from Variable Markups^{*}

Alexander Boca[‡]

University of Michigan

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Abstract

This paper studies how sectoral aggregation affects misallocation estimates in models with variable markups. In a nested CES framework with oligopolistic competition, misallocation depends on markup dispersion within and across sectors, weighted by substitution elasticities. Applying this to European firm-level data reveals two countervailing biases. First, higher aggregation (for example, grouping meat and bakery into a broader food sector) mechanically smooths out dispersion, reducing measured misallocation. Second, aggregation biases down estimated elasticities. The mechanical bias typically dominates: using aggregated sectoral definitions leads to a quantitatively significant underestimation of aggregate productivity losses.

Keywords: Misallocation, Productivity, Variable Markups

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[‡]*E-mail:* aboca@umich.edu.

1 Introduction

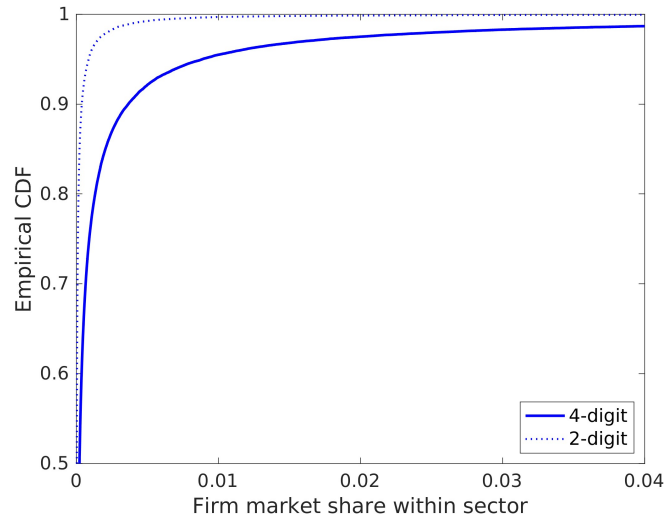
There is a growing interest in the economic literature in measuring the consequences of market concentration for aggregate productivity. A decisive step in this measurement is the definition of the market, or the level at which firms compete. Ideally, markets would be defined based on product substitutability, but for practical reasons researchers in macroeconomics and international economics often instead rely on conventional industrial classifications (e.g., NACE or NAICS). In doing so, they must inevitably determine the level of aggregation at which a market operates. This paper studies the consequences of such reliance, specifically studying how the level of sectoral aggregation affects estimates of misallocation in models with variable markups.

The theoretical framework is the nested CES demand model with oligopolistic competition. This class of models, originating from Atkeson and Burstein (2008), has been widely used to quantify productivity losses from market power (see Edmond et al. (2015), Edmond et al. (2023), Choi et al. (2024)). Its popularity stems from a key advantage: it generates model-consistent markups from price-setting behavior without requiring the direct estimation of production functions. However, while the model is theoretically robust, its empirical application faces a challenge: national data classifications differ in aggregation, and firm-level balance-sheet data often lack fine sectoral definitions. Despite a general notion that aggregated sectoral definitions might underestimate misallocation (Baqee and Farhi, 2020), the literature has not systematically quantified the direction or magnitude of this aggregation bias.

To visually demonstrate this mechanism upfront, Figure 1 displays the empirical cumulative distribution function (CDF) of Spanish manufacturing firms' market shares using both 4-digit and 2-digit sector classifications. Broadening the sectoral definition from 4-digits to 2-digits not only reduces the magnitude of market shares, but, more importantly, compresses their dispersion. In this family of models, dispersion in market shares reflects dispersion in markups, which generates efficiency losses compared to a benchmark with constant markups. Figure 2 shows the impact of this compression, plotting average TFP loss from variable markups across different aggregation levels.¹ As sectoral definition becomes broader, the computed misallocation monotonically declines. The level of aggregation, therefore, matters in a quantitatively large and stable fashion.

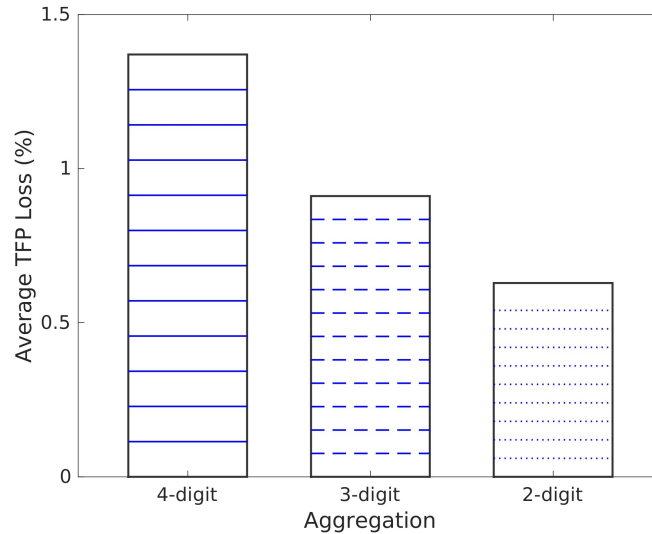
¹Methodological details and the exact calculations for these figures are discussed in Section 4.

Figure 1: Distribution of Market Shares by Sectoral Aggregation



Notes: The figure displays the empirical cumulative distribution function (CDF) of firm market shares for the Spanish manufacturing sector in 2019. Sectors are defined at the 4-digit (solid line) and 2-digit (dotted line) NACE Rev. 2 levels. Broadening the sectoral definition mechanically compresses the dispersion of market shares. Data is sourced from the Orbis Historical database.

Figure 2: TFP Loss by Sectoral Aggregation



Notes: This figure plots the average total factor productivity (TFP) loss due to variable markups for the Spanish manufacturing sector between 2003 and 2019. The TFP loss represents the efficiency gap relative to a benchmark economy with constant markups. Estimates are compared across three levels of NACE Rev. 2 sectoral aggregation: 4-digit, 3-digit, and 2-digit. Consistent with the theoretical mechanism, broadening the sectoral definition compresses the measured dispersion of markups, causing the computed misallocation to decline monotonically. Data is sourced from the Orbis Historical database.

I combine a theoretical decomposition with a quantitative application using comprehensive Spanish manufacturing data. First, to provide intuition, I derive a second-order approximation of the model. Misallocation is driven by the weighted sum of markup dispersion both between and within sectors. Because the model implies a positive, convex relationship between markups and market shares, misallocation depends on the dispersion of sectoral concentration. Intuitively, highly concentrated and competitive sectors are more likely at the granular 4-digit level, i.e., high dispersion in concentration, than at the 2-digit level, where these differences average out. I confirm that this mechanism holds in the data.

Second, I quantify this bias by inverting the model to recover firm-level primitives (relative productivities and prices), following the approach in Choi et al. (2024). This inversion allows me to recover firm- and sector-level productivity directly from observed market shares, accommodating the large number of firms found in balance sheet data. For elasticities of substitution, I rely on estimates from Burstein et al. (2025) and assume a nested-CES world where an econometrician estimates elasticities consistent with the available sectoral definition.

Using data from the Orbis Historical Dataset, which offers a representative sample of the firm size distribution, I find that misallocation significantly decreases with sectoral aggregation. I decompose this finding into a mechanical bias and a parameter bias. First, the mechanical bias: bundling granular sub-sectors into broader categories smooths out market share dispersion, systematically pulling measured misallocation downward. For example, holding elasticities of substitution constant, defining Spanish markets at the 2-digit rather than the 4-digit level drops measured misallocation from 1.4% to 0.6% (average 2003–2019). Note that these numbers are small relative to Hsieh and Klenow (2009)-style exercises because this paper focuses only on the variable markups component. Second, a parameter bias: aggregation alters the estimated elasticities of substitution. While the mechanical bias typically dominates, leading to an underestimation of misallocation, the parameter bias introduces significant sensitivity. In fact, when the gap between estimated elasticities widens sufficiently, this countervailing force can become strong enough to fully offset the mechanical bias or even lead to overcorrection in specific periods.

Finally, because the Orbis Historical dataset requires an exhaustive cleaning process to yield a nationally representative sample, I conduct a battery of robustness checks. These exercises include using operating revenue rather than sales to define market shares, restricting the sample to firms

with more than 50 employees, relaxing balance-sheet filters, and adopting alternative elasticity values common in the literature. Furthermore, I extend the analysis to five other European economies and to non-manufacturing sectors.

The structure of the paper is as follows. Section 2 introduces the nested CES framework, the misallocation measure, and its second-order approximation. Section 3 describes the data and the estimation of elasticities. Section 4 presents the quantitative results on aggregation bias. Section 5 discusses robustness and extensions. Section 6 concludes.

2 Theoretical Framework

2.1 Model

This section describes the nested-CES framework with variable markups, as in Atkeson and Burstein (2008). This specific version follows closely Edmond et al. (2015) in a closed economy.

The final consumption good, Y , is produced by aggregating intermediate inputs y_s sourced from a continuum of sectors indexed by s :

$$Y = \left(\int_0^1 y_s^{\frac{\theta-1}{\theta}} ds \right)^{\frac{\theta}{\theta-1}}, \quad (1)$$

where $\theta > 1$ is the elasticity of substitution between sectors. Within each sector s , there is a finite number of intermediate good producers N_s indexed by i . Output in sector s is produced by aggregating inputs y_{is} .

$$y_s = \left(N_s^{-\frac{1}{\gamma}} \sum_{i \in s} y_{is}^{\frac{\gamma-1}{\gamma}} \right)^{\frac{\gamma}{\gamma-1}}, \quad (2)$$

where $\gamma > 1$ is the elasticity of substitution between goods in a sector. Note that it is assumed that goods within a sector are more substitutable than sector composites, i.e. $\gamma > \theta$.

Each intermediate good producer y_{is} uses labor as sole input:

$$y_{is} = a_{is} l_{is}, \quad (3)$$

where a_{is} is firm productivity. Final good producers maximize profits by choosing optimal quanti-

ties of intermediate inputs:

$$PY - \int_0^1 \left(\sum_{i \in s} p_{is} y_{is} \right) ds \quad (4)$$

subject to the nested demand structure (1)-(2). The demand for intermediate inputs:

$$y_{is} = \left(\frac{p_{is}}{p_s} \right)^{-\gamma} \left(\frac{p_s}{P} \right)^{-\theta} Y, \quad (5)$$

where the aggregate and sectoral price indices, P , and p_s are CES aggregates with elasticities θ and γ , respectively:

$$P = \left(\int_0^1 p_s^{1-\theta} ds \right)^{\frac{1}{1-\theta}}, \quad (6)$$

$$p_s = \left(\frac{1}{N_s} \sum_{i \in s} p_{is}^{1-\gamma} \right)^{\frac{1}{1-\gamma}}. \quad (7)$$

Intermediate goods producers face the demand given by (5) and compete á la Cournot within each sector. They take wages as given and incur a fixed cost of operation, f , paid in units of labor. Then, profits are given by the following function:

$$\max_{y_{is}} \left[\left(p_{is} - \frac{W}{a_{is}} \right) y_{is} - Wf \right], \quad (8)$$

The solution implies the following price:

$$p_{is} = \frac{W}{a_{is}} \frac{\varepsilon_{is}}{\varepsilon_{is} - 1}. \quad (9)$$

Intermediate firms set prices as a markup over marginal cost, with the markup determined by the price elasticity of demand, $\mu_{is} = \varepsilon_{is}/(\varepsilon_{is} - 1)$. Using (9) and under Cournot competition, the price elasticity of demand:

$$\frac{1}{\varepsilon_{is}} = \frac{1}{\gamma} + \left(\frac{1}{\theta} - \frac{1}{\gamma} \right) \omega_{is}, \quad (10)$$

where ω_{is} is the firm's i market share in sector s . Under Cournot competition, the elasticity of demand depends on the firm's market share. Specifically, the elasticity is lower (and the markup higher) for firms with larger market shares. This framework implies that firms with greater mar-

ket shares charge higher markups, introducing firm-level heterogeneity in markups due to market structure. Using the demand for intermediate inputs (5) and the price indices (6) and (7), the market shares can be expressed in terms of prices and indices:

$$\omega_{is} = \frac{p_{is}y_{is}}{\sum_{i \in s} p_{is}y_{is}} = \frac{p_{is}^{1-\gamma}}{p_s^{1-\gamma}} \quad (11)$$

$$\omega_s = \frac{p_s y_s}{\int_0^1 p_s y_s ds} = \frac{p_s^{1-\theta}}{P^{1-\theta}} \quad (12)$$

2.1.1 Markups and Market Shares

Combining the definition of markup with the expression for the price elasticity of demand (10) yields:

$$\frac{1}{\mu_{is}} = 1 - \frac{1}{\gamma} - \left(\frac{1}{\theta} - \frac{1}{\gamma}\right)\omega_{is}, \quad (13)$$

which implies a positive and convex relationship between firm-level markups and market shares.

Similarly, it is possible to define a sectoral markup as $\frac{1}{\mu_s} = \sum_{i \in s} \omega_{is} \frac{1}{\mu_{is}}$:

$$\frac{1}{\mu_s} = 1 - \frac{1}{\gamma} - \left(\frac{1}{\theta} - \frac{1}{\gamma}\right)HHI_s, \quad (14)$$

where HHI_s is the Herfindahl-Hirschman Index of sector s . The strength of this relationship is governed by the difference between the inverse of the elasticities of substitution. Intuitively, a larger gap between θ and γ implies that large firms in a given sector can raise their markups without facing significant competition from firms in other sectors.

An additional implication is that a higher dispersion in market shares translates into a higher dispersion in markups.

2.1.2 Market Clearing and Equilibrium

A firm operates in the economy if its profits exceed the fixed costs of operation, f (in labor units). Entry follows a sequential ordering based on productivity: the most productive firm decides whether to operate assuming no other firm operates; then the second most productive decides to join the market taking into account the presence of the first, and so on.

A representative household consumes the final good and supplies one unit of labor inelastically. Market clearing requires that total labor demand, including fixed costs, equals the inelastic labor supply of 1:

$$\int_0^1 \sum_{i \in s} (l_{is} + f) ds = 1. \quad (15)$$

Additionally, aggregate consumption equals final output ($C = Y$).

The equilibrium is obtained by solving a fixed-point problem in prices. The algorithm starts by setting $\varepsilon_{is} = \gamma$. Given productivity draws and normalizing the wage to 1, prices are computed using equation (9), while equation (11) determines market shares. These shares are then used to update the elasticities ε_{is} via equation (10). This process repeats until convergence.

While the literature commonly assumes a distributional process for firm-level productivity (e.g., Edmond et al. (2015), Edmond et al. (2023)) and estimates parameters by matching moments via simulation, this process is computationally costly and often results in imperfect model fit. Imperfect fit is particularly problematic when measuring aggregation bias, as the residual mismatch may be conflated with the bias itself.

To avoid this, I follow the nonparametric approach in Choi et al. (2024), which inverts the model to recover primitives directly from observed market shares. Combining equations (9), (11), and (13):

$$\omega_{is} = \frac{\mu_{is}^{1-\gamma} a_{is}^{\gamma-1}}{\sum_{j=1}^{N_s} \mu_{js}^{1-\gamma} a_{js}^{\gamma-1}}, \quad (16)$$

where μ_{is} is known from equation (13). For each sector s , this system of $N_s - 1$ equations identifies relative productivities a_{is} by normalizing a reference firm (e.g., the firm with the lowest share) to $a_{fs} = 1$. Analogously, sectoral productivities a_s are recovered by normalizing a reference sector. Finally, prices are derived using equation (9). This procedure ensures the model matches observed shares ω_{is} and ω_s perfectly.

2.1.3 Aggregate Productivity and Markups

Aggregate productivity is defined as the ratio of final output to labor input net of entry costs, $\tilde{L}$:

$$A = \frac{Y}{\tilde{L}}, \quad \text{where} \quad \tilde{L} = \int_0^1 \sum_{i \in s} \frac{y_{is}}{a_{is}} ds. \quad (17)$$

Substituting labor demand yields aggregate productivity as the output-weighted harmonic mean of firm productivity:

$$A = \left[\int_0^1 \sum_{i \in s} \frac{1}{a_{is}} \frac{y_{is}}{Y} ds \right]^{-1}. \quad (18)$$

Similarly, the aggregate markup μ is defined as the ratio of the price index to the wage-adjusted aggregate productivity ($P/(W/A)$):

$$\mu = \left[\int_0^1 \sum_{i \in s} \frac{1}{\mu_{is}} \frac{p_{is} y_{is}}{PY} ds \right]^{-1}. \quad (19)$$

2.2 Misallocation

Using the definitions of sectoral markups $\mu_s = p_s/W/a_s$ and the firm and aggregate production functions, aggregate productivity can be expressed as:

$$A = \left[\int_0^1 \left(\frac{\mu_s}{\mu} \right)^{-\theta} a_s^{\theta-1} ds \right]^{\frac{1}{\theta-1}}, \quad (20)$$

where sectoral productivity a_s is given by:

$$a_s = \left(\sum_{i \in s} \left(\frac{\mu_{is}}{\mu_s} \right)^{-\gamma} a_{is}^{\gamma-1} \right)^{\frac{1}{\gamma-1}}. \quad (21)$$

Variable markups reduce sectoral and aggregate productivity compared to an efficient benchmark (A^{eff}) where markups are uniform ($\mu_{is} = \mu$). This efficient scenario may arise if $\theta = \gamma$.

$$A^{eff} = \left[\int_0^1 (a_s^{eff})^{\theta-1} ds \right]^{\frac{1}{\theta-1}}, \quad (22)$$

where

$$a_{eff} = \left(\sum_{i \in s} a_{is}^{\gamma-1} \right)^{\frac{1}{\gamma-1}} \quad (23)$$

Misallocation is defined as the log difference $\ln(A^{eff}/A)$. To provide intuition into what is driving misallocation, I derive a second-order approximation. First, by combining the definition of sector and aggregate markups (19) with those of sector and aggregate productivity (20-21), it is possible

to derive the following expressions for aggregate and sector-level markups:

$$\mu(s) = \left(\sum_{i=1}^{N(s)} \left(\frac{a_i(s)}{a(s)} \right)^{\gamma-1} \mu_i(s)^{1-\gamma} \right)^{\frac{1}{1-\gamma}} \quad (24)$$

and

$$\mu = \left(\int_0^1 \left(\frac{a(s)}{A} \right)^{\theta-1} \mu(s)^{1-\theta} ds \right)^{\frac{1}{1-\theta}}. \quad (25)$$

Next, taking a second-order expansion of A , a_s , their first-best counterparts, and μ and μ_s around $E[\log(a_{is})]$ and $E[\log(\mu_{is})]$, where expectations are taken within each sector, misallocation is approximated as follows:

$$\ln \left(\frac{A^{eff}}{A} \right) \simeq \frac{1}{2} \gamma E[\text{var}(\ln \mu_{is})] + \frac{1}{2} \theta \text{var}(\ln \mu_s), \quad (26)$$

where expectations $E[\cdot]$ and variances $\text{var}(\cdot)$ are taken across sectors. Misallocation is thus the weighted sum of markup dispersion within sectors (weighted by γ) and across sectors (weighted by θ). See Appendix A for the detailed derivation.

2.3 Effects of Aggregation on Misallocation

The approximation above highlights that misallocation is driven by markup dispersion, which in turn stems from dispersion in market shares within sectors, and cross-sector variation in concentration (equations 13 and 14). Conditional on the elasticities of substitution, greater heterogeneity in market shares translates into greater dispersion in markups and hence larger productivity losses.

Aggregating to broader sectoral classifications (e.g., 2-digit rather than 4-digit) masks within-sector heterogeneity. By compressing the variance of market shares, this aggregation mechanically dampens measured misallocation for a given elasticity pair.

Using a Taylor expansion of the markup function, approximate TFP loss can be expressed in terms of observable sufficient statistics:

$$\text{TFP Loss (\%)} \simeq \frac{\phi^2}{(\alpha - \phi \overline{\text{HHI}}_s)^2} [\gamma E[\text{var}(\omega_{is})] + \theta \text{var}(\text{HHI}_s)], \quad (27)$$

where ω_{is} is the market share of firm i in sector s , $\overline{\text{HHI}}_s$ is the average sectoral HHI, $\alpha = 1 - \frac{1}{\gamma}$, and $\phi = \frac{1}{\theta} - \frac{1}{\gamma}$. Holding elasticities fixed, this expression links approximate misallocation to within-sector dispersion in firm market shares and to cross-sector dispersion in concentration. Because broader sectoral definitions compress both magnitudes, the mechanical effect of aggregation is to lower approximate misallocation.

Aggregation also affects the elasticity estimates. As shown in Section 3.2, both the within-sector elasticity $\hat{\gamma}$ and the across-sector elasticity $\hat{\theta}$ decline with aggregation. The effect of this re-estimation on misallocation is, however, theoretically ambiguous due to two competing forces. On the one hand, lower elasticities reduce the weights on within- and across-sector markup dispersion. On the other hand, they alter markups through ϕ , which can amplify markup dispersion.

Appendix B isolates the elasticity effect by holding fixed the concentration and market-share distributions at a given aggregation level and asking how measured misallocation changes when the elasticity pair is replaced by the one re-estimated at a broader level of aggregation. In the monotone case in which γ , θ , and $\kappa = \phi/\alpha$ all fall, approximate misallocation declines unambiguously. When broader-level elasticities instead imply lower γ and θ but higher κ , however, the sign is not determined and must be resolved quantitatively.

3 Data and Estimation Strategy

3.1 Orbis Historical Database

The Orbis Historical Database provides annual firm-level balance sheet data. While coverage varies by country, sector, and period, it is particularly comprehensive for many European countries; see Kalemli-Özcan et al. (2024) and Bajgar et al. (2020). The historical vintage offers a distinct advantage over the commercial product as it retains firms even if they stop reporting for a certain number of years. This allows for the construction of annual samples that are representative of the universe of the cross-sectional firm-size distribution, including both public and private firms (see Jaimovich et al. (2023)), which is critical for measuring misallocation from variable markups. I access these data through the University of Michigan’s Great Lakes Cluster.

To ensure the raw data accurately represents the economy, I implement the rigorous cleaning

process outlined in Gourinchas et al. (2024). First, to avoid double-counting, I filter the data based on consolidation status, predominantly retaining unconsolidated accounts and restricting each firm to a single, dominant reporting history. Second, I enforce strict balance-sheet consistency by dropping firm-year observations that violate standard accounting identities within a 1 percent tolerance. Third, I eliminate entire firm histories if they report negative values for key variables (total assets, employees, sales, or tangible fixed assets), exhibit unrealistic size-specific year-to-year growth jumps in employment or revenue,² or report more than two million employees. Finally, to further mitigate the effect of outliers, I remove firms in the top 0.5 percentile of the employees-to-assets, employees-to-operating revenue, and operating revenue-to-assets ratios, and exclude any remaining firm-year observations with missing or zero values for sales and total assets.

I employ the NACE Rev. 2 classification to standardize market definitions across European countries. I compute misallocation separately for sections C (Manufacturing), G (Wholesale and Retail Trade), H (Transportation and Storage), and J (Information and Communication). In these sectors, I expect markups to exhibit the direct relationship with market shares predicted by the model. I exclude other service sectors because the geographical dimension of market concentration is likely more relevant for market power. Since Orbis data is at the firm level rather than the establishment level, it is not possible to define market shares at the relevant local level. I also exclude sectors with high government participation, as this likely distorts the relationship between markups and market shares.

The baseline results use sales data from the Spanish manufacturing sector (Section C). Section 5 presents robustness checks examining operating revenue, alternative sample selection criteria, data filters, and elasticity parameters. Furthermore, it extends the analysis to non-manufacturing sectors, utilizes granular country-specific sector classifications, and studies other countries with high revenue coverage in the Orbis database.

Using the Spanish sample from 2003 to 2019, I compute annual firm and sector-level market shares at the 4, 3, and 2-digit NACE Rev. 2 levels. I further compute the moments required for the second-order approximation. These moments include: within-sector market share dispersion and

²Following standard literature thresholds, these are scaled by lagged employment. For example, firms are dropped if employment growth exceeds 1,000% for micro-firms (0–10 employees), scaling down to a 100% threshold for firms with over 100 employees. Similar size-dependent thresholds are applied to revenue growth. If lagged employment is missing, a conservative baseline threshold of 2,000% is applied to revenue growth.

the dispersion of the Herfindahl-Hirschman Index (HHI) across sectors.

3.2 Elasticities of Substitution and Aggregation

To compute the elasticities of substitution at different levels of aggregation, I assume that competition actually occurs at the disaggregated 4-digit sector level (denoted by subscript d), while the econometrician estimates parameters using data aggregated to broader sectoral definitions (denoted by subscript a , e.g., 2-digit or 3-digit)³. This discrepancy introduces aggregation bias, which I detail below.

First, taking logs of the demand for intermediate inputs from equation (5) yields the true firm-level demand:

$$\tilde{y}_i = -\gamma_d \tilde{p}_i + (\gamma_d - \theta_d) \tilde{p}_d + \theta_d \tilde{P} + \tilde{Y}, \quad (28)$$

where the tilde denotes natural logarithms, and $\tilde{p}_d$ is the true price index of the disaggregated sector. The sector-level counterpart implies:

$$\tilde{y}_d = -\theta_d \tilde{p}_d + \theta_d \tilde{P} + \tilde{Y}. \quad (29)$$

Additionally, firms set prices as a markup over marginal cost:

$$\tilde{p}_i = \tilde{\mu}_i + \tilde{m} c_i. \quad (30)$$

A standard identification strategy for an econometrician observing prices, quantities, and cost shocks is to estimate these equations in first differences to eliminate time-invariant heterogeneity. Specifically, one would regress changes in firm-level quantities ($\Delta \tilde{y}_i$) on changes in prices ($\Delta \tilde{p}_i$), including aggregated sector fixed effects (α_d) to control for the sector price index $\tilde{p}_d$:

$$\Delta \tilde{y}_i = -\gamma_d \Delta \tilde{p}_i + \alpha_d + u_{id}. \quad (31)$$

The estimated $\hat{\gamma}_d$ is then used to construct the CES aggregates $\tilde{y}_d$ and $\tilde{p}_d$ at the aggregated level,

³This procedure is analogous to the approach in Head and Mayer (2026), who compare elasticity estimates derived from CES demand against a data-generating process based on Berry et al. (1995), a standard model in the Industrial Organization literature.

which are subsequently used to estimate the across-sector elasticity:

$$\Delta \tilde{y}_d = -\theta_d \Delta \tilde{p}_d + u_d. \quad (32)$$

In this framework, the true data-generating process (DGP) is defined at the disaggregated level d . Estimation at any broader level a suffers from Omitted Variable Bias (OVB). In equation (31), relying on fixed effects for the broad sector a fails to control for the unobserved true fine price index $\tilde{p}_d$, leaving the term $(\gamma_d - \theta_d)\Delta \tilde{p}_d$ in the error structure. The probability limit of the estimator $\hat{\gamma}_a$ is therefore:

$$\text{plim } \hat{\gamma}_a = \gamma_d - (\gamma_d - \theta_d) \frac{\text{Cov}(\Delta \tilde{p}_i, \Delta \tilde{p}_d)}{\text{Var}(\Delta \tilde{p}_i)}, \quad (33)$$

where variances and covariances are computed within the aggregated sector a . Because goods are strictly more substitutable within fine sectors than across them ($\gamma_d > \theta_d$), and firm price changes positively covary with their own narrow sector price index ($\text{Cov} > 0$), it follows that $\hat{\gamma}_a < \gamma_d$. Intuitively, bundling heterogeneous products into a broader basket artificially dampens their apparent substitutability, biasing the estimated within-sector elasticity downward.

Analogously, the bias for the across-sector elasticity θ can be expressed as:

$$\hat{\theta}_a = 1 + (\theta_d - 1) \frac{\Delta \tilde{p}_d}{\Delta \tilde{p}_a}. \quad (34)$$

Since consumers substitute more easily across narrow categories than across broad aggregates, the price index response at the disaggregated level is dampened relative to the aggregated level, implying $\frac{\Delta \tilde{p}_d}{\Delta \tilde{p}_a} < 1$. Consequently, $\hat{\theta}_a < \theta_d$, meaning aggregation also mechanically underestimates the across-sector elasticity.

To quantify this, I proceed by calibration and simulation. First, I calibrate the true parameters $\gamma_d = 5.0$ and $\theta_d = 1.8$ following Burstein et al. (2025) and invert the model using 4-digit NACE Rev. 2 sectors to recover the underlying primitives. I then simulate the economy by subjecting firms to idiosyncratic cost shocks in equation (30). Using this generated data, I run instrumental variable (IV) regressions based on equation (31) with fixed effects at the 3- and 2-digit levels to obtain $\hat{\gamma}_3$ and $\hat{\gamma}_2$. Finally, constructing the respective sectoral outputs and price indices, I estimate

equation (32) to recover $\hat{\theta}_3$ and $\hat{\theta}_2$ ⁴.

For the baseline results that follow, I use the estimates from 2019, reported in Table 1 below. Figure A1 in the Appendix presents the full estimation results, confirming that these parameters remain stable across years.

Table 1: Estimated Elasticities of Substitution under CES Aggregation

Level of Aggregation	$\hat{\gamma}_a$	$\hat{\theta}_a$
4-digit (True DGP, d)	5.000	1.800
3-digit ($a = 3$)	4.704 (0.057)	1.721 (0.028)
2-digit ($a = 2$)	4.495 (0.079)	1.404 (0.133)

Notes: The table presents the estimated within-sector ($\hat{\gamma}_a$) and across-sector ($\hat{\theta}_a$) elasticities of substitution for the Spanish manufacturing sector in 2019. The 4-digit parameters correspond to the calibrated truth in the underlying data-generating process (DGP). Estimates for the broader 3-digit and 2-digit aggregate levels are obtained via instrumental variable regressions on simulated idiosyncratic cost shocks. Standard errors, clustered at the respective sectoral level of estimation, are reported in parentheses. Consistent with the derived omitted variable bias, estimating elasticities at broader levels of aggregation biases the estimates downward. Data is sourced from the Orbis Historical database.

4 Quantifying the Aggregation Bias

This section presents the baseline results for Spain at the 4-digit level, and compares them against calculations defining 3, and 2-digit sectors both when using the same elasticities at the 4-digit level and re-estimating the elasticities. This section also provides intuition for explaining this measurement error based on a second-order approximation.

4.1 Baseline Misallocation

First, I compute misallocation using granular 4-digit sector definitions for the period 2003–2019. The results, along with the second-order approximation, are presented in Figure 3. Misallocation in Spanish manufacturing increased from 1.0% in 2003 to 1.8% in 2019. This upward trend is

⁴As a validation check, applying this procedure back to the baseline 4-digit data perfectly recovers the true parameters γ_d and θ_d .

consistent throughout the sample, with a notable exception during the 2013–2016 period, which corresponds to the recovery phase following the European Financial Crisis.

Misallocation appears to be counter-cyclical with respect to real activity. I find a negative correlation ($\rho = -0.74$) between misallocation and manufacturing real Gross Value Added (GVA). This implies that allocative efficiency improves during booms and deteriorates during downturns, which aligns with theories where recessions reduce competitive entry.

Additionally, misallocation is positively correlated ($\rho = 0.84$) with aggregate market concentration, measured as the sales share of the top-3 firms within each 4-digit sector. As market leaders capture larger shares, markup dispersion widens, mechanically increasing allocative inefficiency.

Finally, Figure 3 confirms that the second-order approximation tracks the exact computed misallocation closely. This approximation allows for a decomposition of the sources of inefficiency. There is a shift in the composition of misallocation over time: from 2003 to 2017, within-sector markup dispersion was the dominant driver (accounting for 57% of the total), whereas in the post-2017 period, between-sector dispersion became the primary source (rising to 55%). This suggests that recent efficiency losses are driven less by markup heterogeneity among direct competitors and more by distortions across different industries.

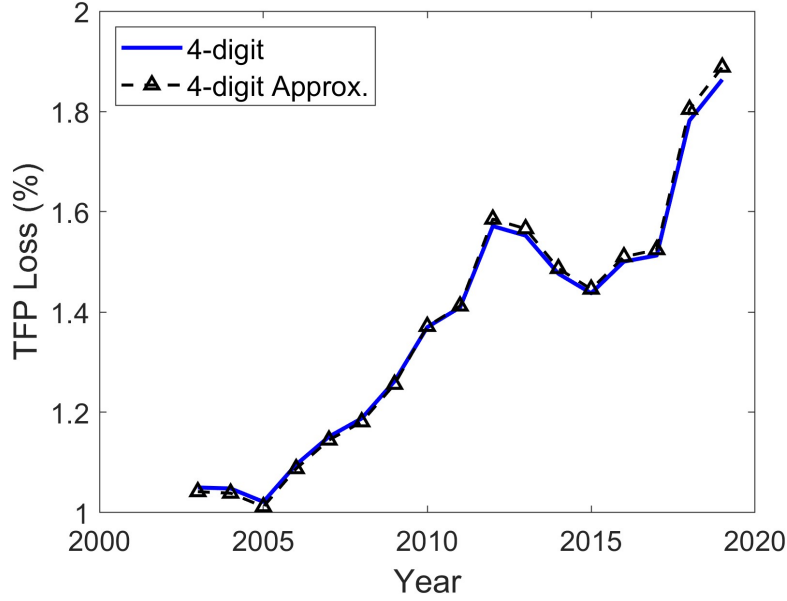
4.2 The Mechanical Bias

Figure 4 presents misallocation calculated at the 3-digit and 2-digit sector levels, while holding the elasticities fixed at their true 4-digit values. This isolates the pure “mechanical” effect of aggregation. In 2019, misallocation drops substantially from 1.8% in the baseline (4-digit) to 1.4% at the 3-digit level, and further to 1.0% at the 2-digit level. This implies that using aggregated data masks almost half of the true allocative inefficiency arising from variable markups.

This exercise is relevant as the literature often uses elasticity estimates from disaggregated studies (e.g., Broda and Weinstein (2006)) but apply them to coarser sectoral data (e.g., because concentration measures are only available at the 2-digit level). The results show that this mismatch leads to a severe downward bias.

The second-order approximation clarifies the mechanism. Since elasticities are held constant in this exercise, the collapse in misallocation is driven entirely by the reduction in the dispersion of market shares and concentration. Figure 5 shows that the terms $E[\text{Var}(\omega_{is})]$ and $\text{Var}(HHI_s)$ in

Figure 3: TFP Loss, Spanish Manufacturing



Notes: This figure plots the evolution of total factor productivity (TFP) loss due to variable markups in the Spanish manufacturing sector from 2003 to 2019. It compares the exact computed misallocation (solid line) with a second-order approximation (dashed line) based on Equation (26). Sectors are defined at the 4-digit NACE Rev. 2 level. The tight fit of the approximation demonstrates that efficiency losses are primarily driven by markup dispersion, validating the use of the approximation for decomposing misallocation into within- and between-sector components. Data is sourced from the Orbis Historical database.

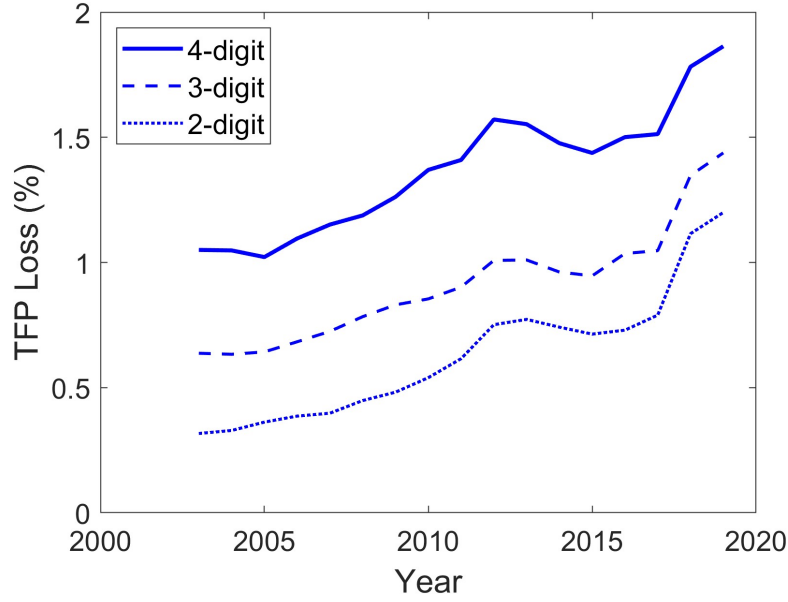
equation (27) shrink with aggregation, mechanically driving down measured misallocation.

Finally, despite the difference in levels, the time-series dynamics remain similar across aggregation levels. The correlation between the 4-digit, and the 3-digit and 2-digit misallocation series is high ($\rho = 0.98$). This finding is similar in spirit to De Ridder et al. (2022), who note that while using revenue (instead of quantity) data biases the level of estimated markups, the trend remains consistent. Similarly, while aggregation leads to a substantial underestimation of the magnitude of efficiency loss, it may have different implications for the dynamics of misallocation.

4.3 The Bias from Elasticities

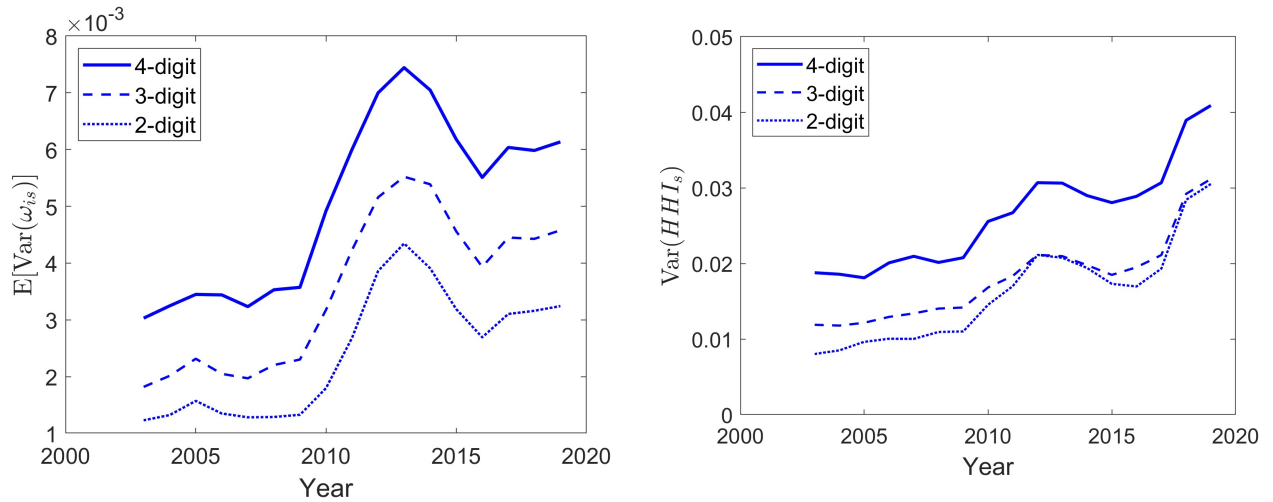
As detailed in Section 3.2, both the within-sector ($\hat{\gamma}$) and across-sector ($\hat{\theta}$) elasticity estimates decrease with aggregation. Section 2.3 showed that, holding fixed the concentration and market-share distributions at a given aggregation level, the effect of this re-estimation on measured misallocation is not signed a priori. Lower elasticities reduce the weights on within- and across-sector markup dispersion, but they may also widen the elasticity gap and raise $\kappa \equiv \phi/\alpha$, thereby amplifying

Figure 4: TFP Loss by Aggregation Level, Constant Elasticities



Notes: This figure plots the aggregate total factor productivity (TFP) loss due to variable markups for the Spanish manufacturing sector from 2003 to 2019 across three levels of NACE Rev. 2 sectoral aggregation (4-digit, 3-digit, and 2-digit). To isolate the pure mechanical effect of aggregation, the within-sector ($\gamma = 5$) and across-sector ($\theta = 1.8$) elasticities of substitution are held constant at their true 4-digit values. The results demonstrate that applying true elasticities to aggregated data mechanically masks markup dispersion, significantly biasing the computed misallocation downward. Data is sourced from the Orbis Historical database.

Figure 5: Dispersion in Market Shares and HHI by Aggregation Level



Notes: This figure plots the components of the second-order approximation of misallocation (Equation 27) for the Spanish manufacturing sector from 2003 to 2019. The left panel shows the expected variance of market shares within sectors ($E[\text{Var}(\omega_{is})]$). The right panel shows the variance of the Herfindahl-Hirschman Index across sectors ($\text{Var}(HHI_s)$). Sectors are defined at the 4-digit, 3-digit, and 2-digit NACE Rev. 2 levels. Broadening the sectoral definition mechanically averages out dispersion both within and across sectors, shrinking these terms and driving down measured misallocation. Data is sourced from the Orbis Historical database.

markup dispersion.

In the Spanish data, the latter force dominates. The estimates in Table 1 imply that κ rises from 0.071 at the 4-digit level to 0.078 at the 3-digit level and 0.109 at the 2-digit level. Quantitatively, this increase in κ more than offsets the direct reduction in the elasticity weights. As a result, replacing the baseline elasticity pair with the one re-estimated at a broader level of aggregation raises measured misallocation conditional on the concentration measures at that level. Thus, in Spain, elasticity re-estimation partially offsets the mechanical downward bias induced by aggregation.

This offsetting effect is not a theoretical necessity. In other countries, the decline in the elasticity weights dominates the increase in markup dispersion, so that applying broader-level elasticities further reduces measured misallocation. Section 5 shows that both cases arise in the data.

4.4 The Total Bias

Figure 6 presents the total effect of aggregation on measured misallocation for Spain, incorporating both the mechanical bias and the re-estimation of elasticities at each aggregation level.

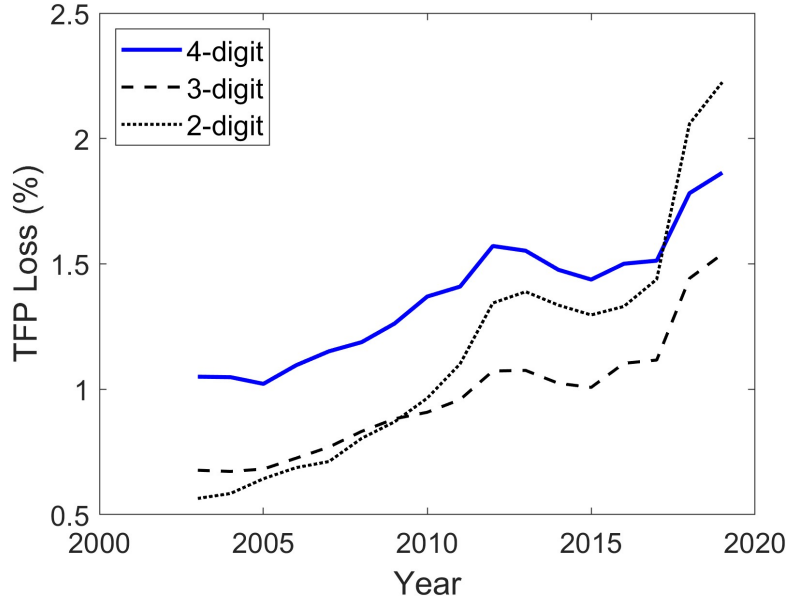
First, despite the upward adjustment from using re-estimated elasticities, the net bias remains negative for most of the sample (2003-2017). Misallocation calculated at the 3-digit and 2-digit levels consistently underestimates the true 4-digit benchmark. This indicates that the loss of granular variance discussed in Section 4.1 dominates the effect from the aggregated elasticities. However, in the final years of the sample (2018-2019), the re-estimated elasticities lead to an overcorrection at the 2-digit level.

Second, the relationship is not uniform across time. After 2009, misallocation at the 2-digit level exceeds that at the 3-digit level. This reversal occurs because the upward effect driven by the re-estimated elasticities outweighs the mechanical drop in markup dispersion when aggregating from the 3- to the 2-digit level during this period (see the right panel of Figure 5).

5 Robustness and Extensions

This section presents alternative specifications to verify the robustness of the baseline results and extends the analysis beyond the core sample. First, I test sensitivity to sample selection criteria, alternative variable definitions, and different elasticity parameters, confirming that the main find-

Figure 6: TFP Loss by Aggregation Level, Re-estimated Elasticities



Notes: This figure plots the total factor productivity (TFP) loss due to variable markups for the Spanish manufacturing sector from 2003 to 2019. It compares misallocation evaluated at the 4-digit, 3-digit, and 2-digit NACE Rev. 2 sectoral levels. To capture the total effect of aggregation, elasticities of substitution are re-estimated at each respective level. For most of the sample period, the mechanical compression of market shares dominates the upward elasticity bias, resulting in a net underestimation of misallocation at coarser levels of aggregation. Data is sourced from the Orbis Historical database.

ings are not driven by data filtering or calibration choices. Next, extending the analysis to other European economies and non-manufacturing sectors confirms that the overall direction of the aggregation bias holds.

5.1 Robustness to Sample Selection and Elasticities

To confirm that the main findings are not artifacts of specific empirical choices, this subsection presents a series of robustness exercises using the baseline Spanish data (extensions to other countries are deferred to the following subsection). I first explore alternative variable definitions and sampling criteria while holding the baseline elasticities constant, and then test the sensitivity of the results to alternative elasticity calibrations.

First, although sales provide a more accurate measure of a firm's market share within its relevant product market, Orbis offers broader coverage for operating revenue, which may include income outside the firm's core business. Figure A3 panel (a) in the Appendix shows that the main result—that TFP loss decreases with aggregation—remains robust to this alternative definition.

Furthermore, to minimize sample attrition, many studies rely exclusively on firms legally obligated to report complete financials. In Spain, firms with over 50 employees face these stricter reporting requirements. Figure A3 panel (b) presents results using this restricted subsample, showing that the findings are qualitatively identical.

Additionally, standard data cleaning procedures can aggressively filter the firm sample. To ensure these restrictions do not drive the main results, I implement an alternative, relaxed cleaning procedure. Specifically, I remove all strict balance-sheet consistency requirements and year-to-year growth-jump outlier exclusions. Figure A3 panel (c) demonstrates that relaxing these raw data filters does not alter the direction of the aggregation bias.

Finally, I test alternative elasticities commonly referenced in the literature, such as $\gamma = 10$ and $\theta = 1.5$ (see Atkeson and Burstein (2008) and Amiti et al. (2019)). Figure A3 panel (d) shows that re-estimating the elasticities at different levels of aggregation and computing TFP loss leads to results qualitatively identical to the baseline. This pair of elasticities, however, leads to higher absolute values of TFP loss.

Across all these alternative specifications, the core conclusion holds: measured misallocation declines with market aggregation.

5.2 Cross-Country Robustness: Belgium, France, Germany, Italy, and Portugal

This section assesses the robustness of the findings by applying the nonparametric approach to five other European economies with good coverage in Orbis.

First, I examine the mechanical effect of aggregation by holding 4-digit elasticities fixed. As shown in Figure A4 in the Appendix, the result is robust across all sampled countries: aggregation reduces measured misallocation.

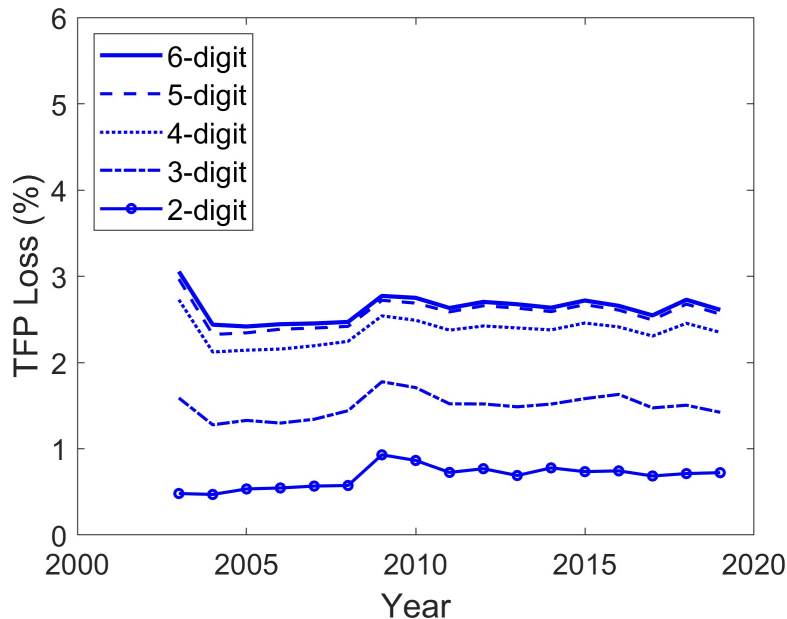
Next, I re-estimate the elasticities at different levels of aggregation across the broader cross-country sample. As shown in Figure A2, while the estimated elasticities ($\hat{\gamma}$ and $\hat{\theta}$) systematically decrease at coarser levels of aggregation, the composite elasticity parameter, κ , exhibits non-monotonic behavior. In most countries, applying these re-estimated elasticities partially offsets the mechanical compression of market share dispersion, thereby yielding an upward correction to

measured misallocation (see Figure A5). However, this countervailing effect is not uniform. For instance, in Belgium at the 3-digit level, the elasticity adjustment also reduces κ and exacerbates the underestimation. This confirms that while re-estimation may act as a countervailing force, the net effect remains theoretically ambiguous.

A potential limitation of NACE is that it harmonizes sectors across countries into standard definitions, potentially smoothing over relevant heterogeneity. To test this, I exploit the granular national classification available for Italy (ATECO), which defines sectors at the 6-digit level.

Figure 7 compares misallocation calculated using the ATECO classification (fixing $\gamma = 5, \theta = 1.8$). Two key results emerge. First, the negative relationship between aggregation and misallocation holds even within this national definition. Second, and more importantly, the level of misallocation at the 6-digit ATECO level is substantially higher than at the 4-digit NACE level (2.6% vs 1.5% in 2019). This suggests that international standardization comes at a cost: by grouping 6-digit national niches into 4-digit European categories, the standard NACE classification masks nearly 40% of the markup dispersion.

Figure 7: TFP Loss by Aggregation Level, Italy's ATECO



Notes: This figure plots the aggregate total factor productivity (TFP) loss due to variable markups for the Italian manufacturing sector from 2003 to 2019. Misallocation is evaluated across five levels of Italy's national ATECO sectoral classification (from 6-digit down to 2-digit). To isolate the pure mechanical effect of aggregation, the within-sector ($\gamma = 5$) and across-sector ($\theta = 1.8$) elasticities of substitution are held constant. The results demonstrate that highly granular 6-digit definitions reveal substantially higher misallocation compared to 4-digit NACE, which masks a significant portion of markup dispersion. Data is sourced from the Orbis Historical database.

5.3 Other Sectors and Asymmetric Aggregation

A key challenge in analyzing the service sector with Orbis data is the lack of establishment-level information. For industries where competition is geographically localized (e.g., restaurants, personal services), national market shares are poor proxies for market power. Consequently, I restrict the analysis to service sectors where markets are plausibly national or broader: Sections G (Wholesale and Retail), H (Transportation and Storage), and J (Information and Communication).

Figure A6 (Appendix) presents the results for these sectors, holding the true 4-digit elasticities constant. While the general pattern of aggregation reducing misallocation holds, Section J (Information and Communication) presents an exception during the 2003–2007 period. Here, moving from the 4-digit to the 3-digit level actually increases measured misallocation.

This anomaly highlights a “classification risk” inherent in statistical definitions. Unlike Manufacturing, which has a mature and balanced hierarchy, newer sectors like Section J often suffer from asymmetric aggregation. When moving up the classification tree, some small 4-digit niches remain independent (becoming 3-digit sectors with high concentration), while others are merged into broad, diluted categories. This process widens the gap between the most and least concentrated sectors. Since misallocation is driven by the dispersion of market shares (not just the average), this asymmetry mechanically inflates the inefficiency measure.

This finding reinforces a critical methodological point: reliance on administrative sector definitions (e.g. NACE, NAICS) introduces non-economic noise. While manufacturing results are robustly smoothed by aggregation, service sectors, particularly those subject to rapid technological change, are sensitive to the arbitrary granularity of statistical classifications.

6 Conclusions

Studies of misallocation arising from variable markups rely on sectoral classifications to define the markets in which firms compete. This paper demonstrates that this choice is not neutral: because misallocation is driven by the dispersion of market shares, the level of aggregation mechanically determines the magnitude of the estimated efficiency loss.

I show that aggregation introduces two biases. First, a mechanical bias: bundling granular sub-

sectors into broader sectors smooths out market share dispersion, which pulls measured misallocation downward. Second, a parameter bias: re-estimating elasticities at higher levels of aggregation changes the effect of these parameters on misallocation. While the mechanical bias generally dominates, leading to underestimation in most periods and countries, the parameter bias introduces significant noise. In some cases (such as Spain in 2017–2019), the re-estimated elasticities can overcorrect, resulting in misallocation measures that exceed the true granular benchmark.

These findings contribute to the literature in two primary ways. First, I quantify the magnitude and direction of the aggregation bias. While standard industrial classifications often mask substantial distortions, the instability of elasticity estimates at aggregated levels cautions against simple correction factors. Second, I provide robust cross-country validation, confirming that this mechanical underestimation of misallocation systematically holds across different European economies, national classification systems, and non-manufacturing sectors.

Ultimately, there is no “correct” statistical level of aggregation. The appropriate market definition depends on the researcher’s stance on where oligopolistic competition actually takes place.

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Appendix

Appendix A: Second-Order Approximation of Misallocation

To provide intuition on the role of sectoral aggregation, I derive a second-order approximation of aggregate misallocation. Given the nested CES demand structure, it is useful to proceed in two steps: first, aggregating firm-level distortions to the sectoral level (the inner nest), and second, aggregating sectoral distortions to the macroeconomic level (the outer nest).

Inner Nest: Within-Sector Aggregation

I begin by examining the inner nest, where firms compete within a given sector s . Define the log-productivity and log-markup of firm i as $t_i(s) = \log(a_i(s))$ and $w_i(s) = \log(\mu_i(s))$, respectively. Their unweighted sectoral averages are denoted as $t = E[\log(a_i(s))]$ and $w = E[\log(\mu_i(s))]$. In a frictionless benchmark where all firms charge identical markups, there is no misallocation. The efficient sectoral productivity is given by the standard CES aggregator:

$$a^{\text{eff}}(s) = \left(\sum_{i \in s} a_i(s)^{\gamma-1} \right)^{\frac{1}{\gamma-1}}, \quad (35)$$

where γ is the elasticity of substitution between firms within the sector. Taking logs, this can be approximated to the second order around the symmetric equilibrium as:

$$\log(a^{\text{eff}}(s)) \simeq t + \frac{1}{2}(\gamma - 1) \text{var}(t_i(s)). \quad (36)$$

However, in the presence of variable markups, actual sectoral productivity $a(s)$ is distorted by the dispersion of relative markups. The actual CES productivity aggregator is defined as:

$$a(s) = \left(\sum_{i \in s} \left(\frac{\mu_i(s)}{\mu(s)} \right)^{-\gamma} a_i(s)^{\gamma-1} \right)^{\frac{1}{\gamma-1}}, \quad (37)$$

where $\mu(s)$ is the sectoral markup index. Taking logs and approximating to the second order, actual sectoral productivity and the sectoral markup deviate from the efficient benchmark as follows:

$$\begin{aligned}\log(a(s)) &\simeq \frac{-\gamma}{\gamma-1}w + t + \frac{1}{2}\frac{\gamma^2}{\gamma-1}\text{var}(w_i(s)) \\ &\quad + \frac{1}{2}(\gamma-1)\text{var}(t_i(s)) - \gamma\text{cov}(w_i(s), t_i(s)) \\ &\quad + \frac{\gamma}{\gamma-1}\log(\mu(s))\end{aligned}\tag{38}$$

and

$$\begin{aligned}\log(\mu(s)) &\simeq -t + w - \frac{1}{2}(\gamma-1)\text{var}(t_i(s)) \\ &\quad + \frac{1}{2}(1-\gamma)\text{var}(w_i(s)) - (1-\gamma)\text{cov}(w_i(s), t_i(s)) \\ &\quad + \log(a(s)).\end{aligned}\tag{39}$$

To isolate the actual sectoral productivity, I substitute the expression for $\log(\mu(s))$ into the equation for $\log(a(s))$. Grouping terms simplifies the expression to:

$$\log(a(s)) \simeq t - \frac{1}{2}\gamma\text{var}(w_i(s)) + \frac{1}{2}(\gamma-1)\text{var}(t_i(s)).\tag{40}$$

Sectoral misallocation is defined as the log-difference between the efficient and actual sectoral productivities. Subtracting the result for $\log(a(s))$ from the benchmark $\log(a^{\text{eff}}(s))$ isolates the efficiency loss, which is driven purely by within-sector markup dispersion:

$$\log\left(\frac{a^{\text{eff}}(s)}{a(s)}\right) \simeq \frac{1}{2}\gamma\text{var}(w_i(s)).\tag{41}$$

Outer Nest: Between-Sector Aggregation

Next, I apply the same logic to the outer nest, aggregating across sectors to find the macroeconomic variables. Define the actual log-productivity and log-markup of sector s as $t(s) = \log(a(s))$ and $w(s) = \log(\mu(s))$. Their respective aggregate unweighted averages are $t = E[\log(a(s))]$ and $w = E[\log(\mu(s))]$. Under constant sectoral markups, the aggregate benchmark productivity A^{eff}

would be defined by the efficient sectoral productivities:

$$\log(A^{\text{eff}}) \simeq t^{\text{eff}} + \frac{1}{2}(\theta - 1) \text{var}(t^{\text{eff}}(s)), \quad (42)$$

where θ is the elasticity of substitution between different sectors, and t^{eff} represents the aggregate average of efficient sectoral log-productivities. Mirroring the inner nest derivation, the actual aggregate productivity A and actual aggregate markup μ are approximated as:

$$\begin{aligned} \log(\mu) &\simeq -t + w - \frac{1}{2}(\theta - 1) \text{var}(t(s)) \\ &\quad + \frac{1}{2}(1 - \theta) \text{var}(w(s)) - (1 - \theta) \text{cov}(w(s), t(s)) \\ &\quad + \log(A) \end{aligned} \quad (43)$$

and

$$\begin{aligned} \log(A) &\simeq \frac{-\theta}{\theta - 1}w + t + \frac{1}{2}\frac{\theta^2}{\theta - 1} \text{var}(w(s)) \\ &\quad + \frac{1}{2}(\theta - 1) \text{var}(t(s)) - \theta \text{cov}(w(s), t(s)) \\ &\quad + \frac{\theta}{\theta - 1} \log(\mu). \end{aligned} \quad (44)$$

Substituting the aggregate markup μ into the expression for A and simplifying yields actual aggregate productivity:

$$\log(A) \simeq t + \frac{\theta - 1}{2} \text{var}(t(s)) - \frac{1}{2}\theta \text{var}(w(s)). \quad (45)$$

Final Result: Aggregate Misallocation

Finally, I compute total aggregate misallocation by taking the log-difference between A^{eff} and A :

$$\log\left(\frac{A^{\text{eff}}}{A}\right) = (t^{\text{eff}} - t) + \frac{\theta - 1}{2}[\text{var}(t^{\text{eff}}(s)) - \text{var}(t(s))] + \frac{1}{2}\theta \text{var}(w(s)). \quad (46)$$

This expression can be simplified using two key properties. First, from the inner nest derivation, the average productivity gap between the efficient benchmark and the actual economy is $(t^{\text{eff}} - t) = \frac{1}{2}\gamma E[\text{var}(w_i(s))]$. Second, because actual sectoral productivity $t(s)$ deviates from efficient sectoral productivity $t^{\text{eff}}(s)$ only by a second-order term, the difference between their variances, $[\text{var}(t^{\text{eff}}(s)) - \text{var}(t(s))]$, is of third-order importance and therefore drops out of the approximation.

Combining these insights yields the final expression for aggregate misallocation:

$$\log \left(\frac{A^{\text{eff}}}{A} \right) \simeq \frac{1}{2} \gamma E[\text{var}(w_i(s))] + \frac{1}{2} \theta \text{var}(w(s)). \quad (47)$$

This confirms that total aggregate misallocation is simply the weighted sum of within-sector markup dispersion and between-sector markup dispersion.

Appendix B: The Elasticity Effect in the Second-Order Approximation

We show that approximate misallocation decreases when γ , θ , and κ all decrease.

Recall the second-order approximation

$$\tilde{L}(\gamma, \theta, \kappa) \equiv \ln \left(\frac{A^{\text{eff}}}{A} \right) \simeq \frac{1}{2} \gamma W(\kappa) + \frac{1}{2} \theta B(\kappa), \quad (48)$$

where

$$W(\kappa) \equiv E[\text{var}_s(\log \mu_{is})], \quad B(\kappa) \equiv \text{var}(\log \mu_s). \quad (49)$$

Using the markup equations,

$$\log \mu_{is} = -\log(\alpha - \phi \omega_{is}) = -\log \alpha - \log(1 - \kappa \omega_{is}), \quad (50)$$

and

$$\log \mu_s = -\log(\alpha - \phi H H I_s) = -\log \alpha - \log(1 - \kappa H H I_s), \quad (51)$$

where

$$\kappa \equiv \frac{\phi}{\alpha}. \quad (52)$$

Define

$$h_\kappa(x) \equiv -\log(1 - \kappa x). \quad (53)$$

Since variances are unaffected by additive constants, we can write

$$W(\kappa) = E[\text{var}_s(h_\kappa(\omega_{is}))], \quad B(\kappa) = \text{var}(h_\kappa(HHI_s)). \quad (54)$$

Now consider any random variable X with support contained in $[0, 1/\kappa_1]$, and define

$$V_X(\kappa) \equiv \text{var}(h_\kappa(X)). \quad (55)$$

We first show that $V_X(\kappa)$ is weakly increasing in κ .

Indeed,

$$\frac{\partial h_\kappa(x)}{\partial \kappa} = \frac{x}{1 - \kappa x}, \quad (56)$$

which is increasing in x for $0 \leq x < 1/\kappa$. Since $h_\kappa(x)$ is also increasing in x , it follows that

$$\frac{d}{d\kappa} \text{var}(h_\kappa(X)) = 2 \text{Cov}\left(h_\kappa(X), \frac{X}{1 - \kappa X}\right) \geq 0, \quad (57)$$

because the covariance of two increasing functions of the same random variable is nonnegative.

Applying this result to $X = \omega_{is}$ within each sector and to $X = HHI_s$ across sectors implies that both $W(\kappa)$ and $B(\kappa)$ are weakly increasing in κ . Therefore, if $\kappa_2 \leq \kappa_1$, then

$$W(\kappa_2) \leq W(\kappa_1), \quad B(\kappa_2) \leq B(\kappa_1). \quad (58)$$

Now consider two parameter triples $(\gamma_1, \theta_1, \kappa_1)$ and $(\gamma_2, \theta_2, \kappa_2)$ such that

$$\gamma_2 \leq \gamma_1, \quad \theta_2 \leq \theta_1, \quad \kappa_2 \leq \kappa_1. \quad (59)$$

Because $W(\kappa) \geq 0$ and $B(\kappa) \geq 0$, the previous inequalities imply

$$\gamma_2 W(\kappa_2) \leq \gamma_1 W(\kappa_1), \quad \theta_2 B(\kappa_2) \leq \theta_1 B(\kappa_1). \quad (60)$$

Adding the two inequalities and multiplying by $1/2$ yields

$$\tilde{L}(\gamma_2, \theta_2, \kappa_2) = \frac{1}{2} \gamma_2 W(\kappa_2) + \frac{1}{2} \theta_2 B(\kappa_2) \leq \frac{1}{2} \gamma_1 W(\kappa_1) + \frac{1}{2} \theta_1 B(\kappa_1) = \tilde{L}(\gamma_1, \theta_1, \kappa_1). \quad (61)$$

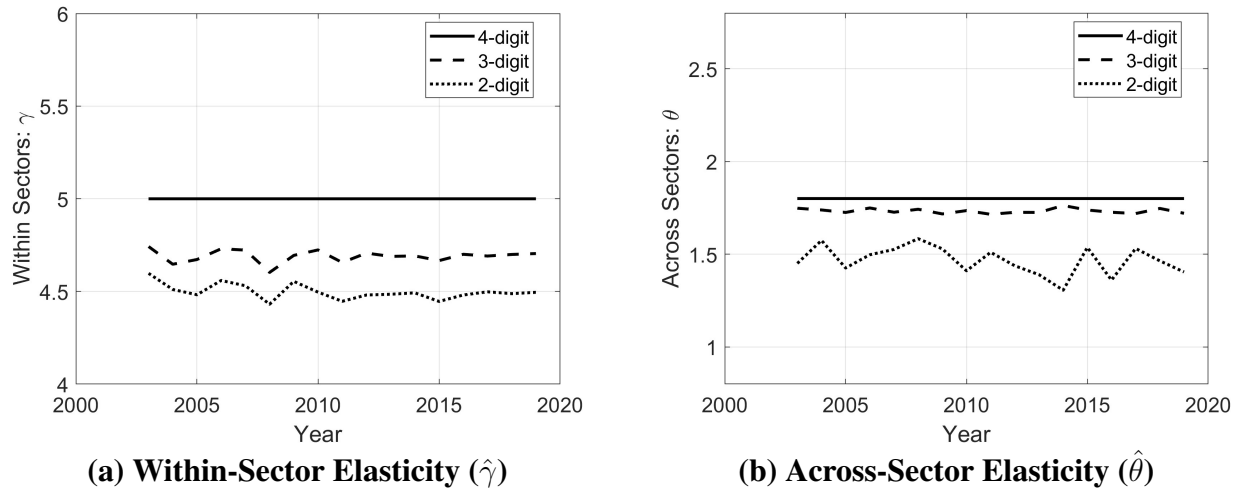
Hence, when γ , θ , and κ all decrease, the second-order approximation of misallocation weakly decreases as well.

The inequality is strict whenever at least one of the parameter inequalities is strict and the corresponding dispersion term is nondegenerate.

Remark. This result applies to the second-order approximation. In the quantitative exercises, that approximation tracks the exact misallocation measure extremely closely; see Figure 3 in the main text.

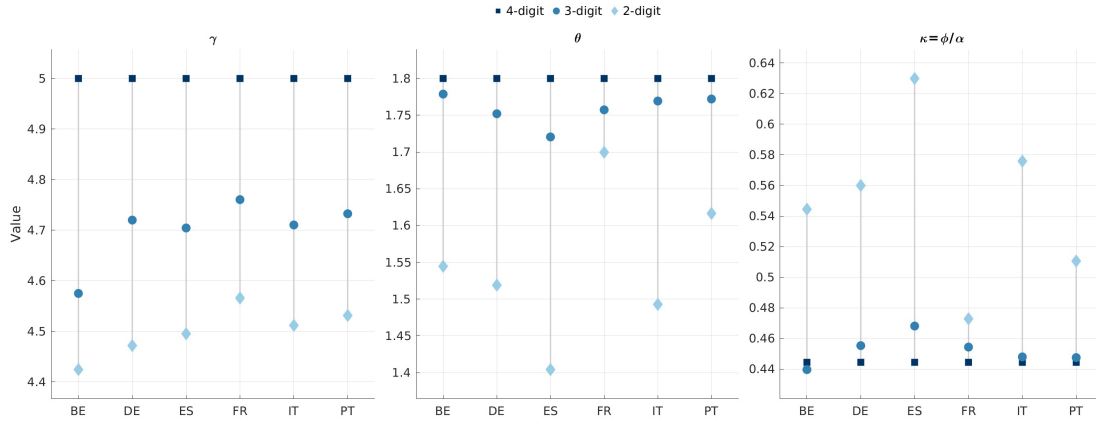
Appendix C: Elasticity Estimation

Figure A1: Estimated Elasticities Across Years, Spanish Manufacturing



Notes: This figure plots the evolution of the estimated elasticities of substitution within sectors (Panel a) and across sectors (Panel b) for the Spanish manufacturing sector over the sample period. Following the procedure in Section 3.2, these elasticities are obtained by simulating the economy with idiosyncratic cost shocks under true 4-digit parameters ($\gamma = 5, \theta = 1.8$) and running instrumental variable regressions on the generated data at the 3-digit and 2-digit levels. This approach demonstrates how aggregating data introduces omitted variable bias, mechanically pushing the estimated elasticities downward. The panels confirm that these downward-biased aggregation estimates remain relatively stable across years.

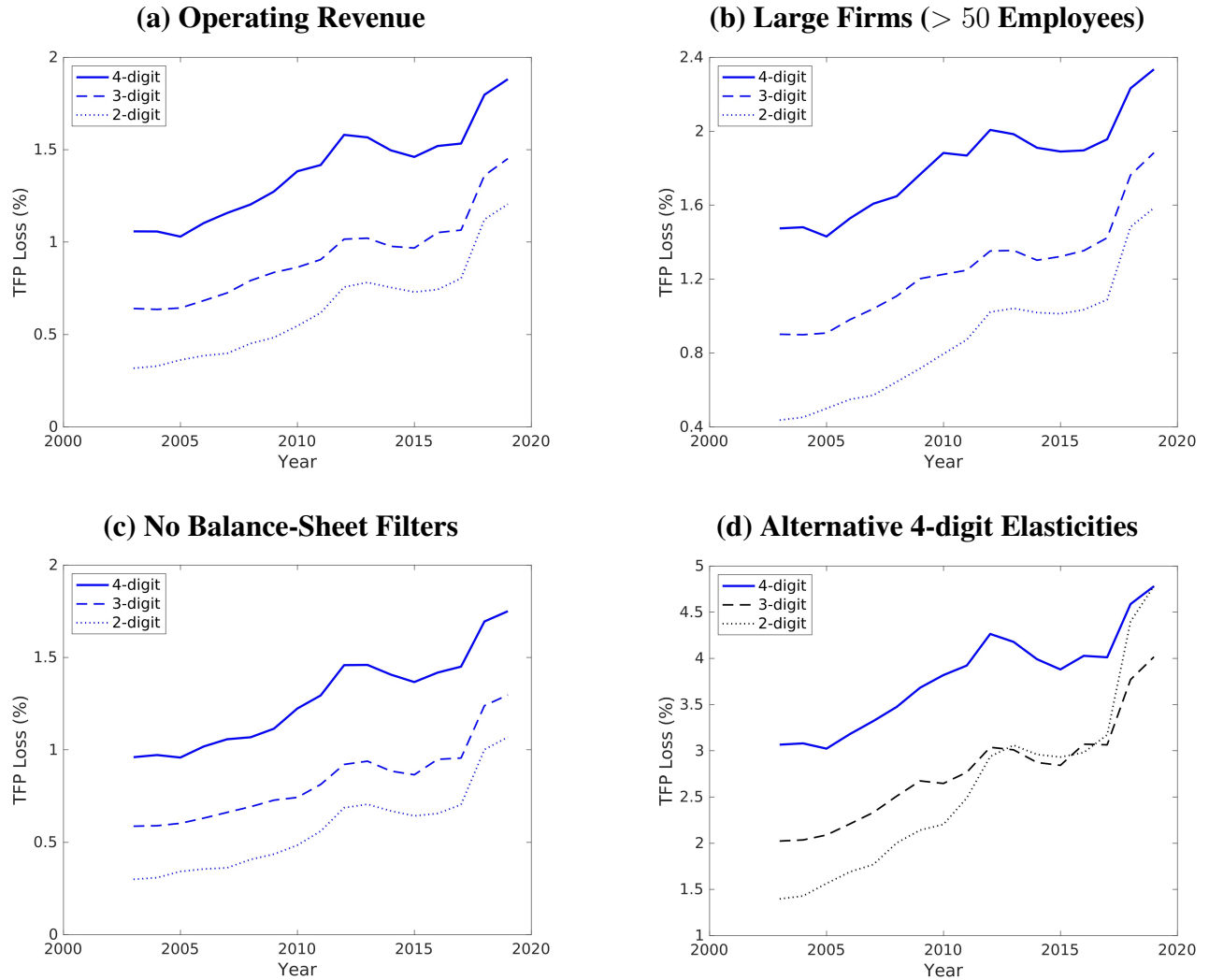
Figure A2: Cross-Country Distribution of Estimated Elasticities



Notes: This figure reports the estimated within-sector ($\hat{\gamma}$) and across-sector ($\hat{\theta}$) elasticities of substitution for 2019 across countries and levels of sectoral aggregation, together with the implied composite elasticity parameter $\hat{\kappa} \equiv \hat{\phi}/\hat{\alpha}$, where $\hat{\phi} = 1/\hat{\theta} - 1/\hat{\gamma}$ and $\hat{\alpha} = 1 - 1/\hat{\gamma}$. As described in Section 3.2, the elasticities are obtained from IV regressions on simulated data designed to recover the bias induced by broad sectoral fixed effects. Consistent with the theoretical framework, both $\hat{\gamma}$ and $\hat{\theta}$ decline as aggregation broadens. The implied response of $\hat{\kappa}$, however, differs across countries. Accordingly, elasticity re-estimation may either offset or reinforce the mechanical downward bias from aggregation. In particular, when broader-level elasticities imply lower $\hat{\gamma}$, lower $\hat{\theta}$, and lower $\hat{\kappa}$, as in Belgium, the elasticity adjustment reduces measured misallocation further.

Appendix D: Robustness Checks

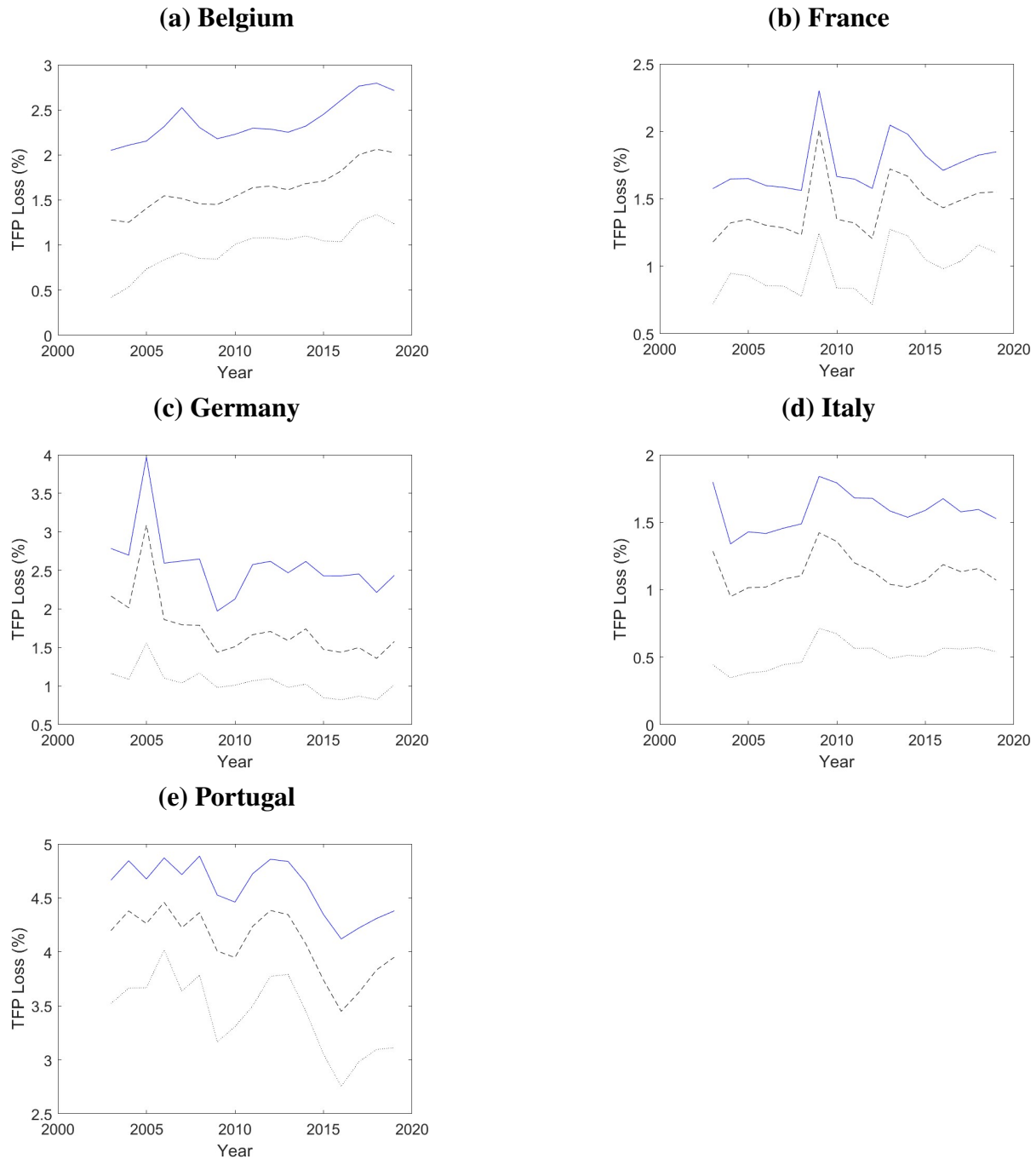
Figure A3: Robustness Checks: Spanish Manufacturing Misallocation by Sectoral Aggregation



Notes: This figure presents robustness checks for the evolution of misallocation driven by variable markups in the Spanish manufacturing sector. Each panel compares the estimated aggregate productivity loss across three levels of NACE sectoral aggregation: 4-digit (solid lines), 3-digit (dashed lines), and 2-digit (dotted lines). To ensure the main findings are not driven by specific data or parameter choices, the panels explore alternative specifications: Panel (a) uses operating revenue instead of sales to define market shares; Panel (b) restricts the sample to large firms (over 50 employees); Panel (c) retains observations without applying balance-sheet identity filters; and Panel (d) assumes alternative true 4-digit elasticities of substitution ($\gamma = 10$ and $\theta = 1.5$). Across all alternative specifications, the core result holds: estimating misallocation at broader levels of aggregation systematically biases the aggregate productivity losses downward.

Appendix E: Cross-Country Baseline Misallocation

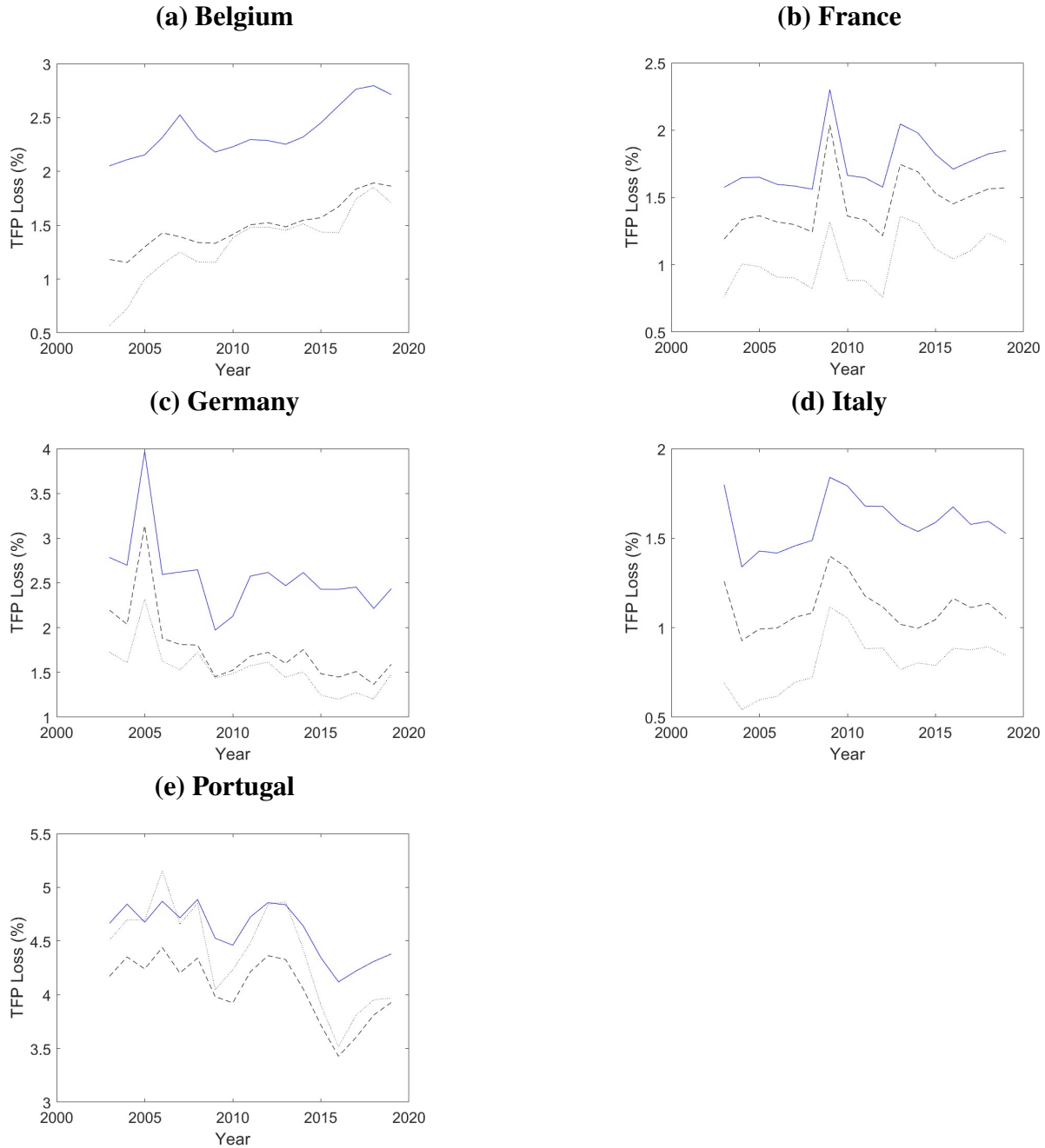
Figure A4: Cross-Country Baseline Misallocation by Sectoral Level of Aggregation



Notes: This figure plots the evolution of misallocation in the manufacturing sector due to variable markups for five European economies. Data is sourced from the Orbis Historical database. Misallocation is computed using the baseline nonparametric approach. Each panel compares the estimated aggregate productivity loss across three levels of NACE sectoral aggregation: 4-digit (solid lines), 3-digit (dashed lines), and 2-digit (dotted lines). Consistent with the main findings, higher levels of aggregation mechanically reduce the measured dispersion of markups, leading to a systematically lower estimate of misallocation across all countries.

Appendix F: Cross-Country Total Misallocation

Figure A5: Cross-Country Total Misallocation by Sectoral Level of Aggregation

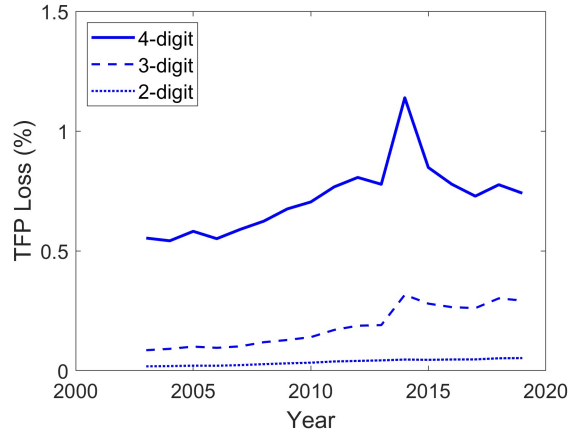


Notes: This figure plots the evolution of total aggregate misallocation driven by variable markups for the manufacturing sector in five European countries. Each panel compares the estimated aggregate productivity loss across three levels of NACE sectoral aggregation: 4-digit (solid lines), 3-digit (dashed lines), and 2-digit (dotted lines). Unlike the baseline estimates which hold substitution parameters constant, these results incorporate the re-estimated within- and across-sector elasticities ($\hat{\gamma}$ and $\hat{\theta}$) specific to each level of aggregation, as detailed in Appendix C. In most cases, these re-estimated elasticities act as a counter-vailing force that partially offsets the mechanical downward bias caused by aggregation. However, this effect is theoretically ambiguous and sensitive to national sectoral clustering (e.g., 3-digit re-estimation in Belgium actually exacerbates the underestimation).

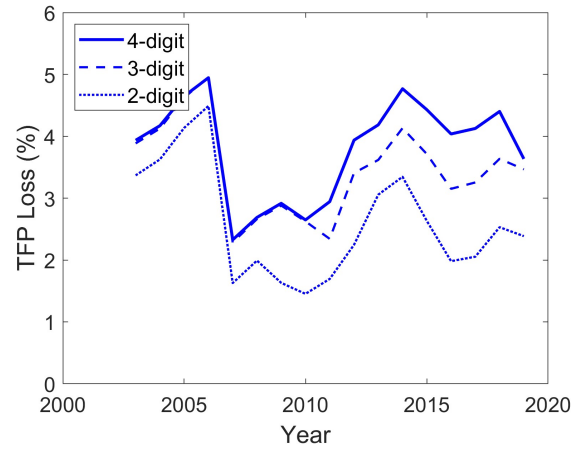
Appendix G: Misallocation in Other Sectors

Figure A6: Baseline Misallocation, Other Sectors in Spain

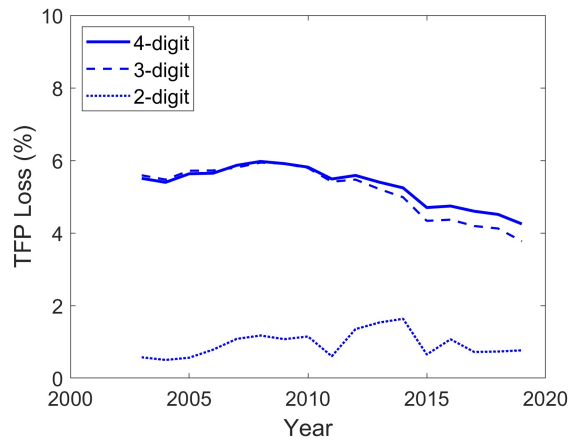
(a) Section G: Wholesale and Retail Trade



(b) Section H: Transportation and Storage



(c) Section J: Information and Communication



Notes: This figure plots the evolution of misallocation driven by variable markups for three non-manufacturing sectors in Spain (NACE Rev. 2 Sections G, H, and J) operating in plausibly national markets. Each panel compares the estimated aggregate productivity loss across three levels of sectoral aggregation: 4-digit (solid lines), 3-digit (dashed lines), and 2-digit (dotted lines), holding the baseline substitution elasticities constant ($\gamma = 5$ and $\theta = 1.8$). While the general pattern of aggregation dampening measured misallocation holds for Sections G and H, Panel (c) highlights an anomaly in Section J during 2003–2007, where 3-digit aggregation actually increases measured misallocation. This stems from asymmetric aggregation (i.e., classification risk) inherent in newer sectors, where merging uneven niches artificially widens the gap between highly concentrated and diluted categories, mechanically inflating the inefficiency measure. This demonstrates how reliance on administrative sector definitions can introduce non-economic noise.